

CONCEPT NOTE · IBM AI PROJECT · 2026



AeroScan

AI-Powered PDF Analysis & Document Intelligence Platform

AI MODEL

LLaMA 3.3 70B

API PROVIDER

Groq Cloud

DOCUMENT

Concept Note



01 · PROJECT IDENTITY

Project Title & Application Name

Project Title: *AeroScan – AI-Powered PDF Analyzer*

Application Name: *AeroScan*

AeroScan is a full-stack web application designed to transform how users interact with PDF documents. The name "AeroScan" evokes speed, precision, and intelligent scanning — reflecting the application's ability to instantly parse, classify, and converse with any text-based PDF document using state-of-the-art large language model technology.

02 · PROBLEM STATEMENT

Problem Statement & Objective

In today's information-heavy world, professionals across industries — finance, legal, healthcare, academia, and logistics — routinely receive dense PDF documents: invoices, contracts, research papers, technical reports, and manuals. Extracting meaningful information from these documents is time-consuming, error-prone, and cognitively demanding.

Traditional approaches require users to manually read, search, and summarise documents — a process that can take hours for a single complex file. Existing solutions either require expensive enterprise licenses, lack conversational capabilities, or fail to provide contextual intelligence.

Core Objectives of AeroScan:

- Eliminate manual document reading by providing instant AI-generated structured summaries of any uploaded PDF.

- Enable users to ask natural-language questions about a document and receive precise, context-aware answers in real time.
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- Automatically classify document types (Invoice, Resume, Contract, Research Paper, etc.) and extract critical entities such as dates, amounts, names, and reference numbers.
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- Deliver all of the above through an accessible, zero-setup web interface requiring no technical expertise from the end user.



03 · AUDIENCE

Target User & Use Case

AeroScan serves a broad spectrum of users who regularly work with structured and semi-structured PDF documents:



BUSINESS PROFESSIONALS

Finance teams, procurement officers, and operations managers who process invoices, purchase orders, and financial reports at scale.



LEGAL & COMPLIANCE TEAMS

Lawyers and paralegals who need to quickly identify key clauses, dates, and parties within lengthy contracts and agreements.



RESEARCHERS & STUDENTS

Academics, graduate students, and data analysts who need rapid comprehension of research papers, technical manuals, and academic publications.



HEALTHCARE PROFESSIONALS

Clinicians and administrators who need to quickly navigate medical reports, patient summaries, and clinical trial documents.

Primary Use Cases:

- **Invoice & Receipt Processing:** Automatically extract totals, vendor names, due dates, and line items from financial documents.
- **Contract Review:** Identify parties, effective dates, termination clauses, and key obligations without reading the full document.

- **Research Summarisation:** Get a structured bullet-point summary of a 50-page research paper within seconds.
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- **Interactive Document Q&A:** Ask specific questions like "What is the total amount due?" or "Who are the signatories?" and get instant answers.

04 · AI INFRASTRUCTURE

LLM Model & API Used

AeroScan leverages cutting-edge large language model infrastructure via the Groq Cloud API:

COMPONENT	TECHNOLOGY	ROLE IN APPLICATION
LLM Model	LLaMA 3.3 70B Versatile	Core AI model powering both document analysis and conversational Q&A. Developed by Meta AI, this 70-billion-parameter model offers best-in-class instruction-following and context understanding.
Inference API	Groq Cloud API	Ultra-fast LLM inference platform using proprietary LPU™ (Language Processing Unit) hardware. Delivers near-instantaneous responses, critical for a smooth user experience.
SDK	groq-sdk (Node.js)	Official Groq JavaScript SDK used in the Express backend to interface with the chat completions API endpoint.
Response Format	JSON Object Mode	Structured JSON responses enforced for the analysis endpoint, ensuring reliable parsing of document type, summary, entities, and confidence score.
PDF Parsing	pdf-parse (Node.js)	Extracts raw text from uploaded PDF binary buffers before sending to the LLM. Supports text-based PDFs up to 10 MB.

Why Groq + LLaMA 3.3? Groq's LPU hardware delivers inference speeds up to 10× faster than GPU-based solutions, enabling AeroScan to analyse a 50,000-character document in under 3 seconds — making it feel instantaneous to the end user. LLaMA 3.3 70B's large context window and strong reasoning capabilities make it ideal for document comprehension tasks.



05 · FEATURES

Key Features of AeroScan

- **Drag-and-Drop PDF Upload:** Users can upload PDF files by dragging and dropping onto the upload zone or clicking to browse. Files up to 10 MB are accepted. Instant visual feedback is provided during the upload and analysis process via a full-screen animated loading state.
- **Intelligent Document Classification:** AeroScan automatically classifies uploaded documents into categories such as Invoice, Receipt, Resume, Contract, Technical Report, Manual, Research Paper, or Unknown — enabling users to understand document context at a glance.
- **AI-Generated Structured Summary:** A concise, bullet-point markdown summary of the document's core content is generated by the LLM and rendered in the Summary tab. Confidence score and document type are displayed as labelled badges for quick reference.
- **Key Entity Extraction:** Critical identifiers are automatically extracted and displayed in a visual card grid — including dates, names, monetary amounts, reference numbers, company names, contact information, and other document-specific fields.
- **In-Document Conversational Chat (Q&A):** Users can ask natural-language questions about the document's content directly in the chat interface. The AI uses the full PDF text as contextual knowledge to answer accurately, stating clearly when information is not available in the document.
- **Inline PDF Viewer:** The original PDF is rendered in a side-by-side iframe panel so users can visually reference the source document while reviewing the AI analysis or chatting — all within one unified workspace.
- **Server-Side Document Caching:** Parsed PDF text and analysis results are cached in-memory on the backend (up to 50 documents) using a unique file session ID. This eliminates redundant re-parsing and re-analysis on each chat interaction, significantly reducing latency.
- **Shared Serverless API Key Model:** A single Groq API key is managed server-side via environment variables. Users require no personal API key or account — making AeroScan accessible to all users

without friction.

- **Dark Theme Premium UI:** A polished dark-mode interface with purple and cyan accent gradients, smooth animations, and responsive layout built with React and vanilla CSS — delivering a premium visual experience.

06 · EXPERIENCE & OUTCOMES

Expected User Experience & Outcomes

End-to-End User Journey:

- 1 Upload**

User lands on a clean drag-and-drop interface. They drop a PDF (e.g. a supplier invoice). The app validates the file and shows the filename with a prominent "Analyze Document" button.
- 2 Analysis**

On clicking Analyze, a full-screen animated loading screen appears while the backend extracts PDF text via pdf-parse and sends it to Groq's LLaMA 3.3 70B model for structured analysis.
- 3 Results**

Within 2–4 seconds, the workspace splits: the original PDF appears on the left; the AI analysis panel appears on the right showing document type, confidence score, and a structured summary.
- 4 Explore Entities**

The user switches to the Key Entities tab to view a card grid of extracted data points — e.g. "Invoice Total: \$4,250.00", "Due Date: 2026-08-01", "Vendor: Acme Corp."
- 5 Chat & Ask**

The user opens the Document Chat tab and types: "Does this invoice include VAT?" — the AI replies with a precise, document-grounded answer. Multiple follow-up questions can be asked in a continuous conversation.
- 6 Reset & Repeat**

User clicks Reset to return to the upload screen and immediately begin analyzing a new document — zero page reloads required.

Expected Outcomes:**Time Savings**

Reduce document comprehension time from hours to under 10 seconds for typical business documents.

**Accuracy**

LLaMA 3.3 70B delivers high-precision entity extraction and contextually accurate Q&A responses grounded in the source document.

**Accessibility**

No sign-up, no API keys for users, no installation — works entirely in a web browser for anyone worldwide.

**Productivity**

Users can process multiple documents in sequence, dramatically increasing throughput for document-heavy workflows.

**Comprehension**

Structured summaries and entity cards provide an at-a-glance understanding that goes beyond simple keyword search.

**Interactivity**

Conversational Q&A turns passive documents into interactive knowledge sources, empowering data-driven decisions.